

ECM3412 - Report

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Introduction

I have implemented an elitist ACO algorithm in Python. My implementation uses OOP and relies on the ACOGraph, Edge and Ant classes that I have defined. These store the relevant attributes and provide the objects with the required methods to allow the simulation to function.

My evaluation has been done using the trial function in main.py. This runs 'n' number of simulations (on a new ACOGraph object each time) and plots the results on the same graph. The Figures 1 through 9 plots with iteration on the x axis, and the global best fitness on the y axis. This shows the algorithms progress over time.

Question 1

I experimented using various permutations of the parameters, number of ants(n), q and the evaporation rate (e). During these tests, I kept α , β , the heuristic and the maximum number of fitness evaluations constant. These were 1, 2, 'transition rule' and 10,000 respectively.

By carrying out a number of tests and storing results in a csv file, I identified the best combination of parameters for the Burma dataset to be n=50, q=0.25 and e=0.25. Over 5 trials, the best fitness using these parameters was 3359. The results for these parameters over 5 trials is in 1(a). It shows that 3 of the 5 results produced results below 3600, and generally converged before 25 iterations.

The best combination of parameters was different for the Brazil dataset. These were n=50, q=0.5 and e=0.5. The use of these parameters produced the best fitness of 27502. This can be seen in 1(b). This not only converged quickly, but produced a result with the lowest fitness.

Question 2

These findings were impacted by the use of randomness implemented within my code. Randomness was used to set the initial pheromone of edges between 0 and 1. It was also used to place each ant at a random starting vertex.

I believe that the reason that these parameters produced the optimal solutions is due to the fact that they contain a balance of q and e, to ensure that good solutions are weighted just right, and pheromone is not evaporated too aggressively, which would mean more used edges would have a stronger impact for more iterations. By using 50 ants, there would be more iterations before reaching 10,000 fitness evaluations compared to more ants, this would mean pheromone is updated more often which enables the best fitness to converge faster.

Question 3

The number of ants is one of the main parameters of an ant colony optimisation(ACO) implementation. It has an effect on the number of iterations that the ACO would run within a given number of fitness evaluations. If a lower number of ants is chosen, then there would be fewer fitness evaluations in an iteration, and as a result, more iterations would be run. This is important in my 'elitist' ACO as pheromone for the best global solution is updated at the end of an iteration, and therefore this would have more impact on the final solution. 2 and 3 show my experiments when altering the number of ants as the independent variable.

The value of q is also impactful in an ACO. It is a constant and is used to weight the quality of a solution. In my implementation, pheromone is updated using (q/ant fitness). A greater value for q would therefore allow a better solution to have more of an impact on the pheromone compared to a less optimal solution. The results when q is altered can be seen in 4 and 5.

The evaporation rate(e) is responsible for evaporating pheromone after each iteration. In my implementation, the pheromone on an edge is multiplied by $(1-e)$. This means that a greater value of e , would mean that more pheromone evaporates after each iteration. As you may see in 6 and 7, a greater value of e sometimes makes the algorithm take longer to converge, which can be a good thing when controlled to allow the ant colony to explore more options.

α and β are two parameters that alter the balance of the transition rule heuristic. A small value for α means that there is less of an importance on the pheromone on an edge with respect to the global pheromone deposits. A smaller value for β will mean that there is less of an importance on the edge cost, with respect to the total cost of edges globally. Typically, α is set to 1, and β is set to 2 to allow the algorithm to try shorter edges.

Question 4

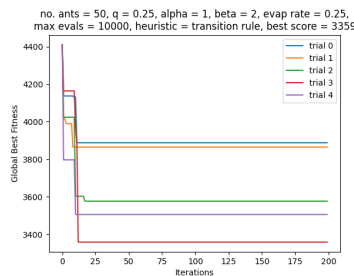
The local heuristic function is used when evaluating the score of an edge. This impacts the decision making process of an ant when choosing the next path. My implementation uses the transition rule, as well as $1/d$, and q/d , the performance of these can be seen in 8 and 9. I feel that a local heuristic function that could be added is one that includes probability. This would allow paths with low pheromone to be taken, which could open up the doors to new solutions and prevent sub-optimal convergence.

Question 5

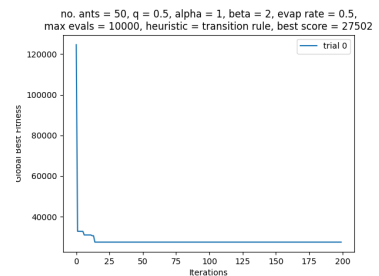
During my implementation, I included an option to run a 'normal' ACO, or an 'elitist' ACO. My experiments and figures have all be obtained using my implementation for an 'elitist' ACO due to its superior ability to yield better results. Another implementation method which could improve my results is a 'MMAS' ACO. This limits the values of pheromone with maximum and minimum bounds. It also changed how pheromone is updated and it only updates pheromone for the best-ant within an iteration. I feel that this could improve my results as by limiting pheromone, convergence would take longer, this would allow more paths to be explored, and could prevent the ant colony reaching a local optimum.

Question 6

The travelling salesman problem(TSP) is a computationally hard problem, especially when scaled up. Bee Colony Optimisation is another nature inspired algorithm that could be applied to the TSP. Bee Colony Optimisation represents solutions as food sources, and bees can be classified into different groups. These different groups evaluate a solution by a different metric and would be very powerful in getting a better fitness for solutions. BCO would therefore require a fewer iterations to converge to an optimal solution.



(a) Burma



(b) Brazil

Figure 1: Best Parameter Combination for Burma and Brazil

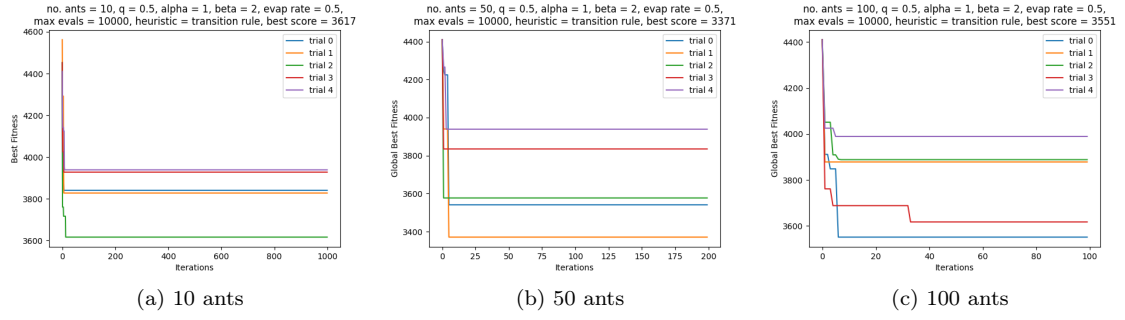


Figure 2: Altering the Number of Ants in Burma

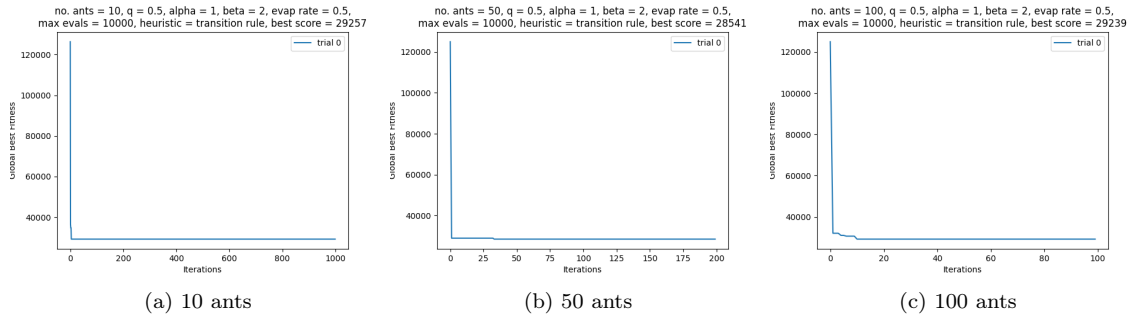


Figure 3: Altering the Number of Ants in Brazil

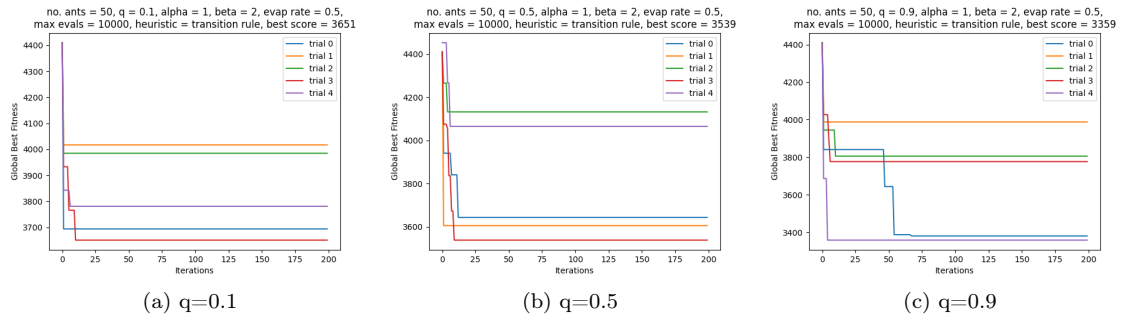


Figure 4: Altering the value of q in Burma

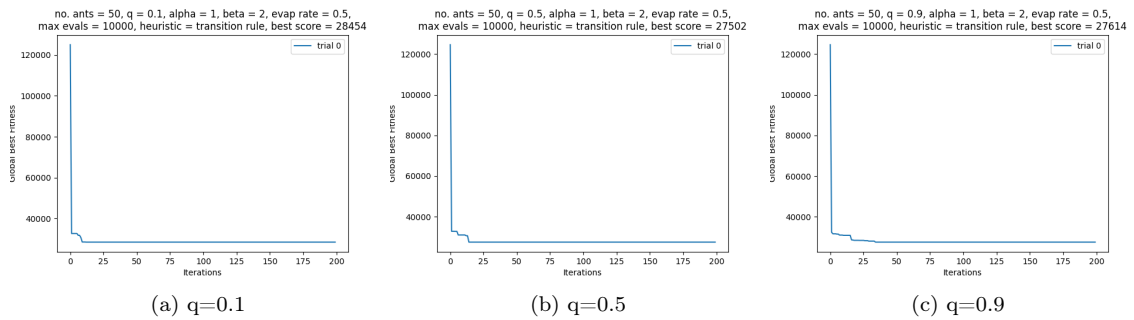


Figure 5: Altering the value of q in Brazil

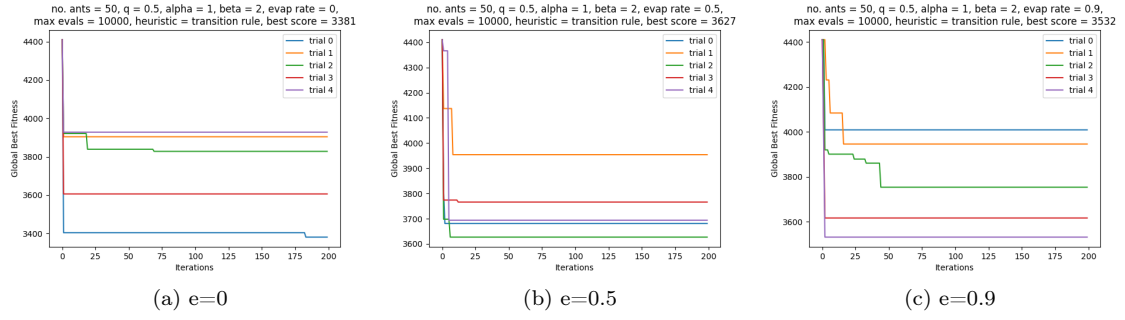


Figure 6: Altering the value of e in Burma

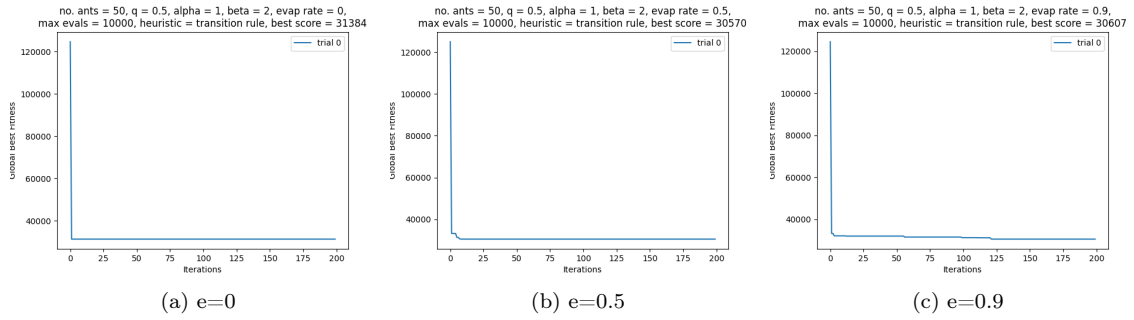


Figure 7: Altering the value of e in Brazil

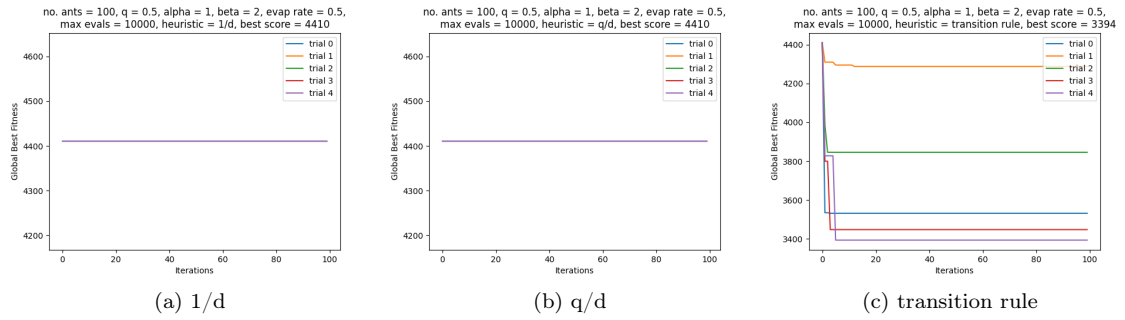


Figure 8: Altering the heuristic in Burma

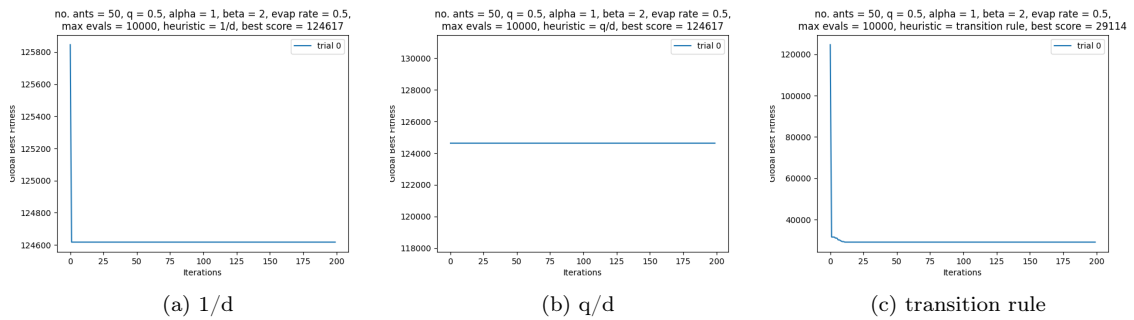


Figure 9: Altering the heuristic in Brazil